Lidar Mapping Report

Acquisition, Processing, and Delivery of Airborne Lidar Elevation Data for Eagleview NYC Project

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Glossary of Terms

Term	Description
AGL	Above Ground Level
AGPS	Airborne Global Positioning System
AGNSS	Airborne Global Navigation Satellite System
ANPD	Aggregate Nominal Pulse Density
ANPS	Aggregate Nominal Pulse Spacing
ASPRS	American Society of Photogrammetry and Remote Sensing
AT	Aerial Triangulation
CD	Compact Disk
CMS	Certified Mapping Scientist
CORS	Continuous Operating Reference Station
CP	Certified Photogrammetrist
CRS	Coordinate Reference System
CVA	Consolidated Vertical Accuracy
DEM	Digital Elevation Model
DFIRM	Digital Flood Insurance Rate Maps
DPA	Defined Project Area
DSM	Digital Surface Model
DTM	Digital Terrain Model
DVD	Digital Versatile Disk / Digital Video Disk
DXF	Data Exchange Format / Drawing Interchange
FIRM	Flood Insurance Rate Maps
FEMA	Federal Emergency Management
FGDC	Federal Geographic Data Committee
FVA	Fundamental Vertical Accuracy
FY	Fiscal Year
GIS	Geographic Information System
GISP	Geographic Information System Professional
GNSS	Global Navigation Satellite System
GPS	Global Positioning System
GSD	Ground Sample Distance
HARN	High Accuracy Reference Network
HDD	Hard Drive Disk
HPGN	High Precision Geodetic Network
IMU	Inertial Measurement Unit
INS	Inertial Navigation System
LAS	(or .las) – industry accepted LIDAR data exchange file format
LB	License Business
LS	Land Surveyor
LiDAR	(or Lidar) Light Detection And Ranging
MARS®	Eagleview Advanced Remote Sensing
Eagleview	Eagleview-Surdex Joint Venture, LLP
MSL	Mean Sea Level
NAD	North American Datum
NDEP	National Digital Elevation Program
NGP	National Geospatial Program
NGS	National Geodetic Survey
NMAS	National Map Accuracy Standards

No.	Number
NPS	Nominal Point Spacing
NSRS	National Spatial Reference System
NSSDA	National Standard for Spatial Data
NVA	Non-vegetated Vertical Accuracy
OPUS	Online Positioning User Service
PDOP	Positional Dilution Of Precision
PLS	Professional Land Surveyor
PLSS	Public Land Survey System
ppsm	Points (or pulses) per square meter
PSM	Professional Surveyor and Mapper
QL2	Quality Level One
QL2	Quality Level Two
RLS	Registered Land Surveyor
RGB	Red, Green, Blue (i.e., three-band image)
RGBNIR	Red, Green, Blue, Near Infra-Red (i.e., four-band image)
RMSE	Root Mean Square Error
SBET	Smoothed Best Estimated Trajectory
SHA	Secured Hash Standard
SPCS	State Plane Coordinate System
SVA	Supplemental Vertical Accuracy
TIN	Triangular Irregular Network
USGS	United States Geological Survey
VVA	Vegetated Vertical Accuracy
WP_ID	Work Package ID (USGS)
WU_ID	Work Unit ID (USGS)
XML	Extensible Markup Language

Project Summary

Eagleview was awarded the NYC Project to provide high resolution data set of 12 points per square meter (ppsm) lidar of approximately 423 square miles over New York City, NY.

The lidar mapping requirements and deliverables meet or exceed a vertical accuracy of 9.25cm RMSEz and a point density of 12ppsm.

Project CRS (Coordinate Reference System)

- Projection – New York State Plane, Long Island
- Horizontal Datum - North American Datum of 1983 (NAD 83), National Adjustment of 2011 (NA2011) (epoch 2010.00)
- Vertical Datum – North American Vertical Datum of 1988 (NAVD 88); using the latest NGS-approved geoid (i.e., **GEOID18**) for converting ellipsoid heights to orthometric elevations
- Horizontal Units – US Feet
- Vertical Units – US Feet

CONTACT INFORMATION

Questions regarding this report should be addressed to:

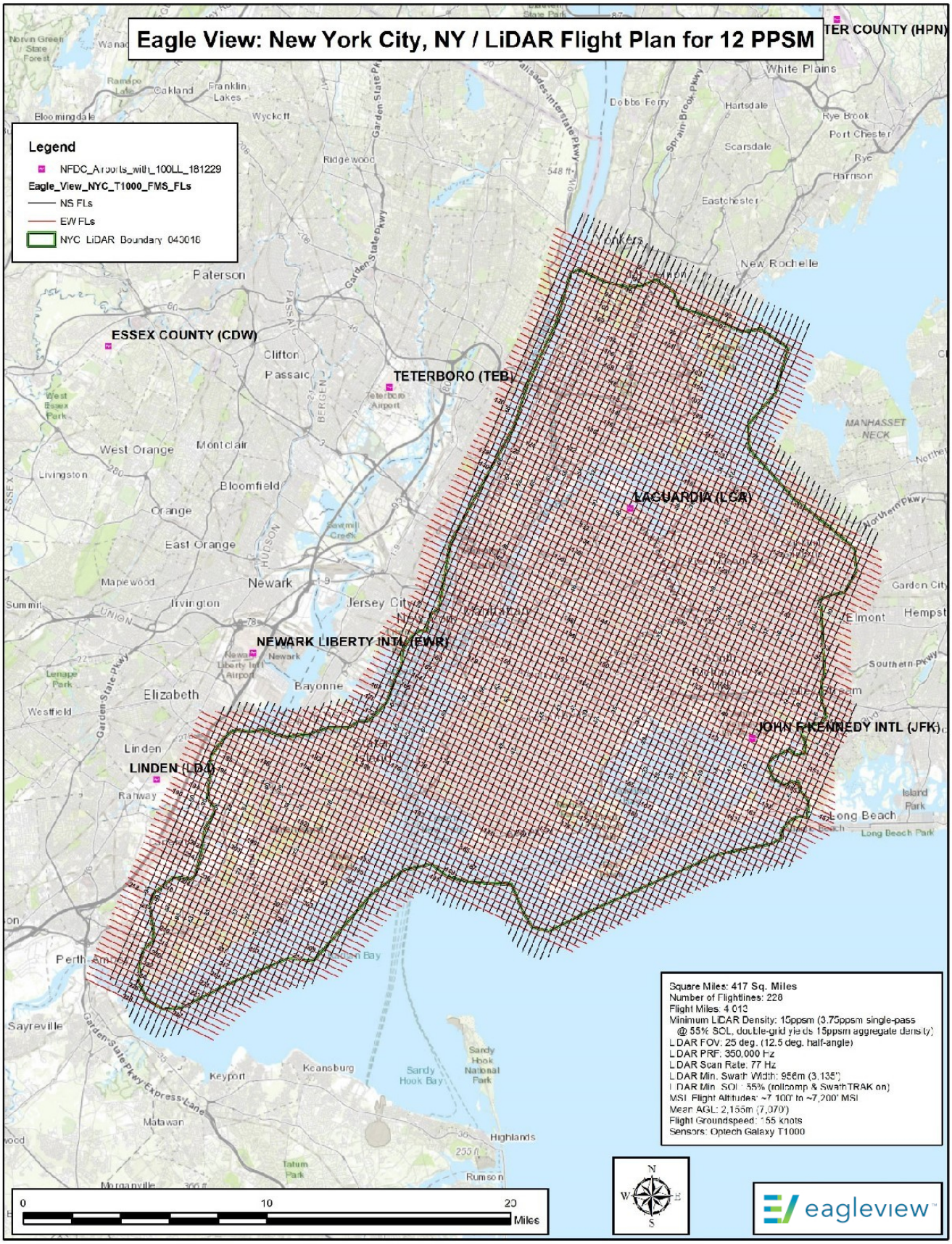
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Project Report

The contents of this report summarize the methods used to calibrate and classify the lidar data as well as the results of these methods for the Eagleview NYC Project.

Lidar Flight Information

The acquisition area or Defined Project Area (DPA) for the Eagleview NYC Project is delineated by the extent of the client approved Esri shapefile (*200m_buff_NYNEWY_Grid_LiDAR.shp*). Eagleview acquired the lidar point cloud utilizing an Optech Galaxy T2000 lidar sensor.


Eagle View: New York City, NY / LiDAR Flight Plan for 12 PPSM
Legend
NFDC_Airports_with_100LL_181229
Eagle_View_NYC_T1000_FMS_FLs
NS FLs
EW FLs
NYC LiDAR Boundary 043018
ESSEX COUNTY (CDW)
TETERBORO (TEB)
LACUARDIA (LGA)
NEWARK LIBERTY INTL (EWR)
JOHN F KENNEDY INTL (JFK)
LINDEN (LDJ)
Square Miles: 417 Sq. Miles
Number of Flightlines: 228
Flight Miles: 4,013
Minimum LiDAR Density: 15ppsm (3.75ppsm single-pass @ 55% SOL, double-grid yields 15ppsm aggregate density)
LiDAR FOV: 25 deg. (12.5 deg. half-angle)
LiDAR PRF: 350,000 Hz
LiDAR Scan Rate: 77 Hz
LiDAR Min. Swath Width: 956m (3,135')
LiDAR Min. SOL: 55% (rollcomp & SwathTRAK on)
MSL Flight Altitudes: ~7,100' to ~7,200' MSL
Mean AGL: 2,155m (7,070')
Flight Groundspeed: 155 knots
Sensors: Optech Galaxy T1000
0
10
20
Miles
N
W
E
S
eagleview

Aerial Mission(s)

Lidar acquisition was collected using fixed wing aircraft and an Optech Galaxy T2000 lidar sensor staging from a variety of airports around the project area. Lidar data collection was accomplished between May 12, 2021 and May 21, 2021. Each mission represents a lift of the aircraft and system from the ground, collects data, and lands again. Multiple lifts within a day are represented by Mission A, B, C, and D. The table below relates each mission to the date collected, the sensor and serial number used, and the actual average MSL in meters.

Mission(s)	Date	Sensor S/N	Actual Avg. MSL (m)
210512_A	May 12, 2019	5060448	2350
210513_B	May 13, 2019	5060448	2365
210513_C	May 13, 2019	5060448	2360
210514_A	May 14, 2019	5060448	2355
210514_B	May 14, 2019	5060448	2355
210515_A	May 15, 2019	5060448	2360
210517_A	May 17, 2019	5060448	2355
210518_A	May 18, 2019	5060448	2365
210518_B	May 18, 2019	5060448	2350
210519_A	May 19, 2019	5060448	2355
210520_A	May 20, 2019	5060448	2355
210520_B	May 20, 2019	5060448	2350
210521_A	May 21, 2019	5060448	2355

GNSS / IMU Data

A five-minute IMU initialization is conducted on the ground, with the aircraft engines running, prior to flight, to establish fine alignment of the IMU. In air IMU calibration maneuvers were performed at the beginning and ending of all mission collections to ensure the best forward and reverse trajectory processing using the highest quality IMU calibration. During the data collection, the operator recorded information on log sheets which includes weather conditions, lidar operation parameters, and flight line statistics. Data is sent back to the main office for preliminary processing to check overall quality of GNSS / IMU data and to ensure sufficient overlap between flight lines. Any problematic data may be reflown immediately as required.

The airborne GNSS data was post-processed using Applanix POSPac Mobile Mapping Suite version 8.x. A fixed-bias carrier phase solution was computed in both the forward and reverse chronological directions. Whenever practical, lidar acquisition was limited to periods when the PDOP was less than 4.0. PDOP indicates satellite geometry relating to position. Generally, PDOP's of 3.0 or less result in a good quality solution, however PDOP's between 3.0 and 5.0 can still yield good results most of the time. PDOP's over 6.0 are of questionable results and PDOP's of over 7.0 usually result in a poor solution. Usually as the number of satellites increase the PDOP decreases. Other quality control checks used for the GNSS include analyzing the combined separation of the forward and reverse GNSS processing from one CORS station and the results of the combined separation when processed from two different CORS stations. An analysis of the number of satellites, present during the flight and data collection times, is also performed.

The GNSS trajectory was combined with the raw IMU data and post-processed using POSPac Mobile Mapping Suite version 8.x. The SBET and refined attitude data are then utilized in the Optech LMS lidar processing software to compute the laser point-positions. The trajectory is combined with the laser range measurements to produce lidar point cloud data.

GNSS Controls

Virtual Ground GNSS Base Station(s) were used to control the lidar airborne flight lines. Post processed Trimble CenterPoint® RTX™ correction service is a high-accuracy, satellite-delivered global positioning service. This technology provides high accuracy GNSS positioning without the use of traditional reference station based differential RTK infrastructure and delivers very high cm level accuracy. In addition, CORS are at times used to further QC or enhance the airborne GNSS solution.

Acquisition Data Check

Validation of field data is a time-critical process. Since re-mobilizations have significant financial and schedule impacts, the JV's goal for every project is to ensure that all data has been completely and accurately acquired before leaving the project site. While coverage is one aspect to verify, the JV focuses on checking aspects that prove adherence to all lidar base specification requirements as well as a full data integrity check as well. Using the MARS® QC Module, the following tests are performed on each mission:

Test	Methodology	Purpose
Returns	Tabular stats and graphics	To ensure all return collecting system components are working properly.
Intensity	Tabular stats and graphics	To ensure all intensity collecting system components are working properly. Also, to look for potential, but rare, laser return path misalignment system issues.
Density	Density calculations by swath but also by spot location, binary raster, density raster, project aggregate, and Voronoi density reporting	To ensure the minimum required lidar point density is achieved for every flight line.
Data Void	Binary raster method as required by LBS	To ensure no unallowable data voids are present
Spatial Distribution	Binary raster method as required by LBS	To ensure all swaths have been collected with the appropriate spatial distribution requirement
Relative Accuracy	Flightline separation raster	An initial look at interswath accuracy, prior to full calibration, to ensure there are no severe and unexpected calibration issues
Sensor Calibration	Scan direction 1 vs 2 separation raster and channel to channel separation raster if applicable	An initial look at intraswath accuracy, prior to full calibration, to ensure there are no severe and unexpected calibration issues
Flight Line Coverage	Coverage rasters	To ensure full coverage of the project boundary. This is a second but different look for data voids.
Sensor Anomalies	Shaded relief raster	To ensure there are no sensor anomalies visible in a shaded relief raster

Lidar Calibration – see appendix 1 for a more detailed workflow description

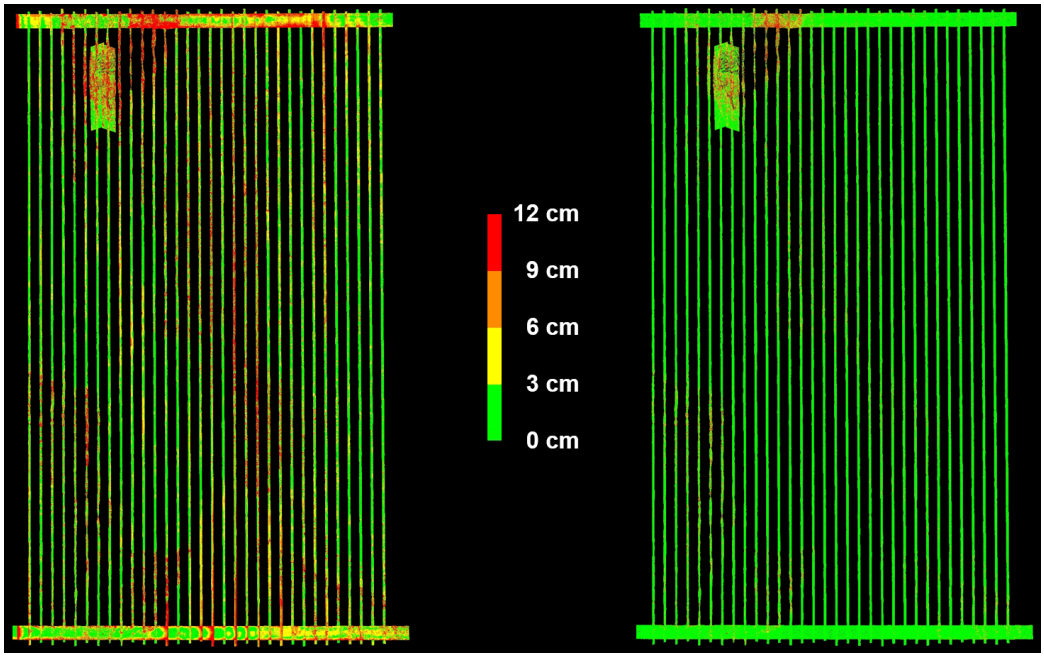
Eagleview takes great care to ensure all lidar acquisition missions are carried out in a manner conducive to post-processing an accurate dataset. Proper Airborne GNSS surveying techniques are always followed including pre- and post-mission static initializations. In-air IMU alignments (figure-eights) are performed both before and after on-site collection to ensure proper calibration of the IMU accelerometers and gyros.

A minimum of one cross-flight is planned throughout the project area across all flightlines and over roadways where possible. The cross-flight provides a common control surface used to remove any vertical discrepancies in the lidar data between flightlines. The cross-flight is critical to ensure flightline ties across the project area. The areas of overlap between flightlines are used to boresight (calibrate) the lidar point cloud to achieve proper flightline to flightline alignment in all three axes. Each lidar mission flown is accompanied by a hands-on boresight in the office.

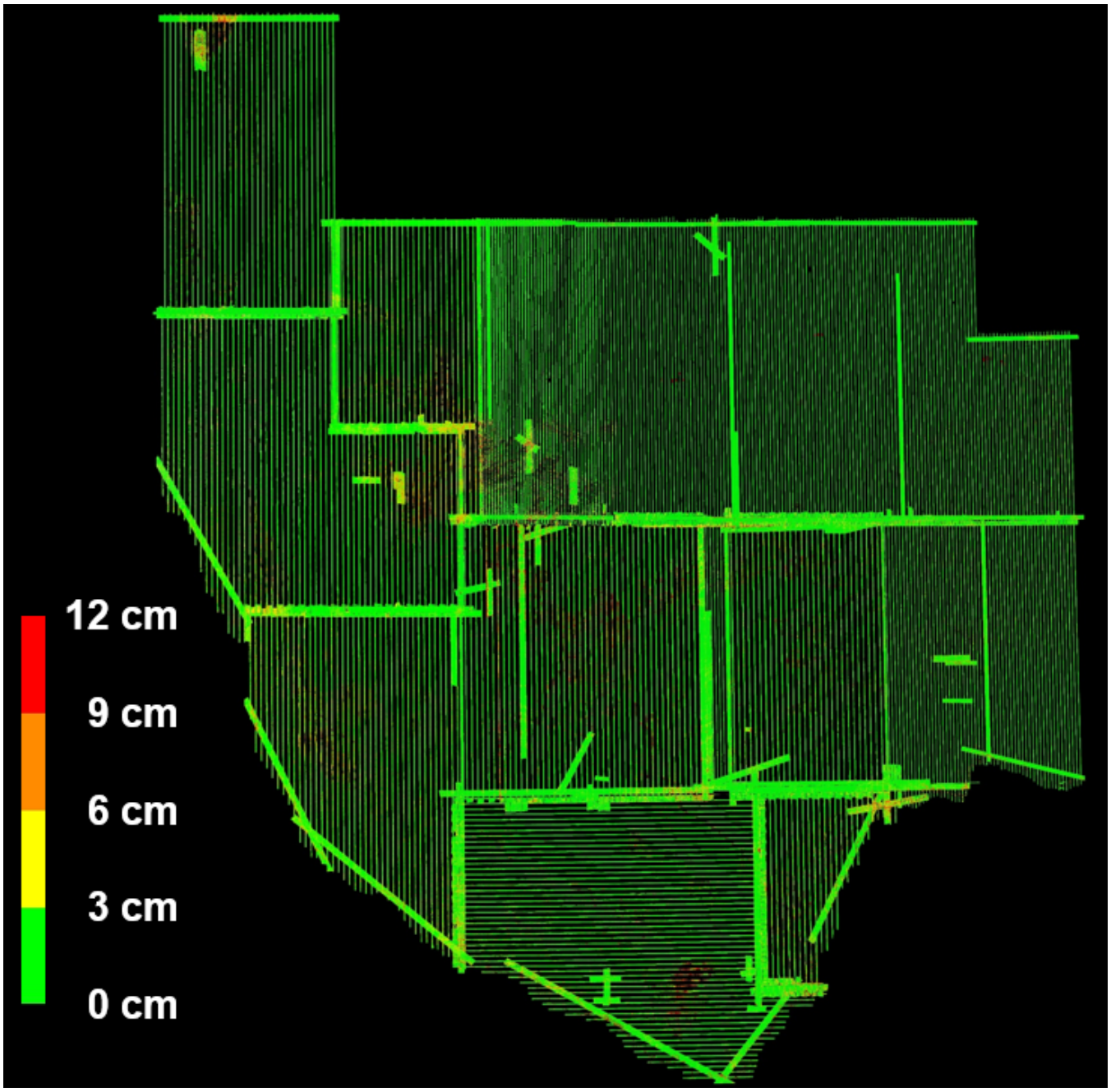
Eagleview understands that high accuracy/quality data cannot be generated from black-boxed-processed lidar data. Many parts of the downstream process suffer from poorly calibrated lidar data. We have a proven process that produces data that meets relative and absolute accuracy specifications reserved for QL0 data for all quality level products. Our all-encompassing lidar calibration process includes the following steps:

1. Sensor model calibration (scale, edge curl, range offsets, etc.)
2. Application of timing offsets (POS and scanner)
3. Calibrating scan direction 0 versus scan direction 1 (inbound versus outbound if applicable)
4. Channel to channel calibration (if applicable)
5. IMU to scanner misalignment angles (heading, pitch, roll deltas) calibration
6. Final geometric calibration tweaks including:
    a. easting, northing, elevations shifts
    b. heading, roll, pitch shifts
    c. easting, northing, elevations drifts
    d. heading, roll, pitch drifts
    e. fluctuating elevation

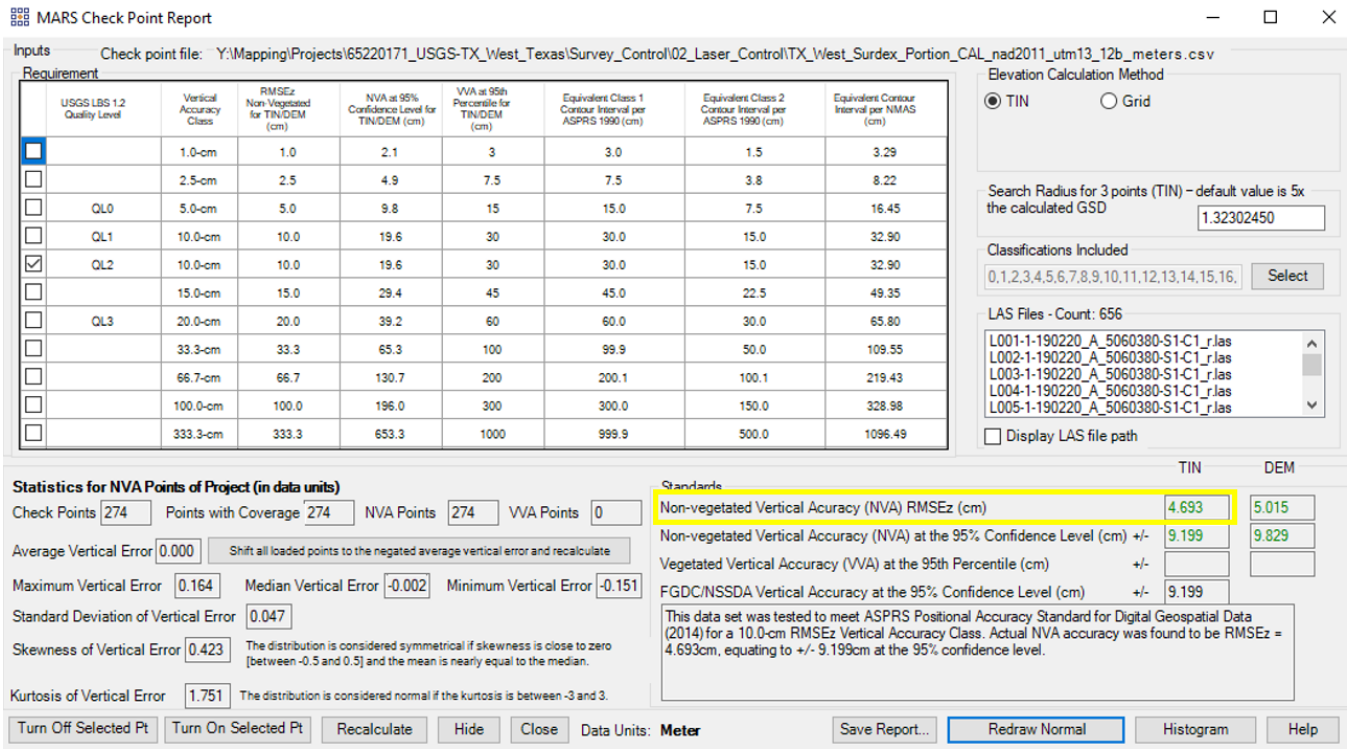
Below is an example of before (left) and after (right) flightline separation rasters having been through this highly effective process. The remaining non-green colors are areas of steep terrain.



Project wide results are equally as accurate.

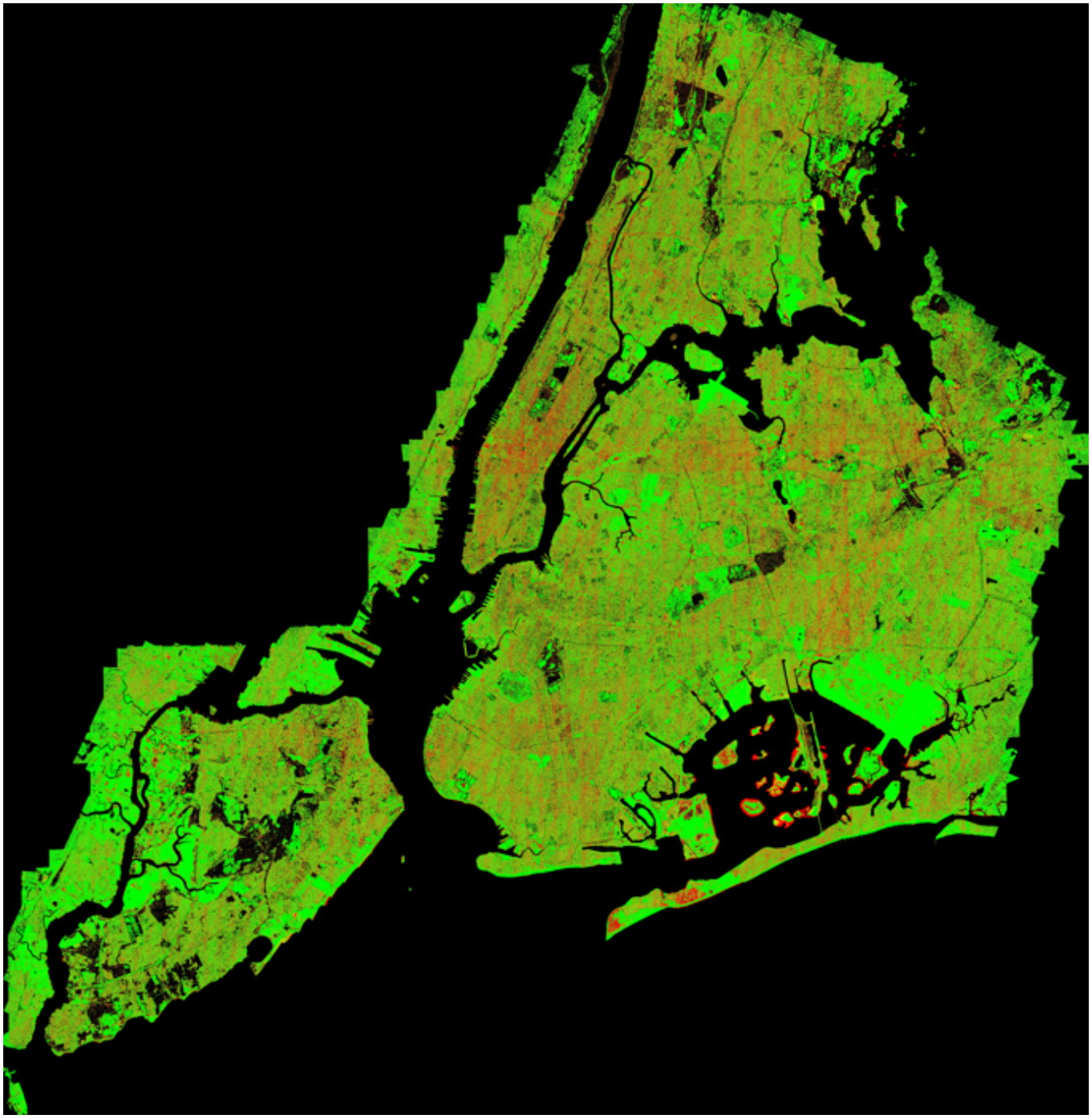

12 cm
9 cm
6 cm
3 cm
0 cm

After boresighting is complete a detailed statistical report is generated to check relative and absolute accuracies before filtering of lidar begins. The calibration process yields excellent absolute accuracies, as can be seen for this example project.



Relative Accuracy – flight line to flight line

The project representative flight line separation raster (below) depicts the vertical separation of flight lines by thematically coloring the separation magnitude on a color ramp based on relative distance.


0
0.080
0.160
0.240
0.800 (Meter)
0
0.262
0.525
0.787
2.625 (Feet)

Unfiltered Lidar Control Point Report

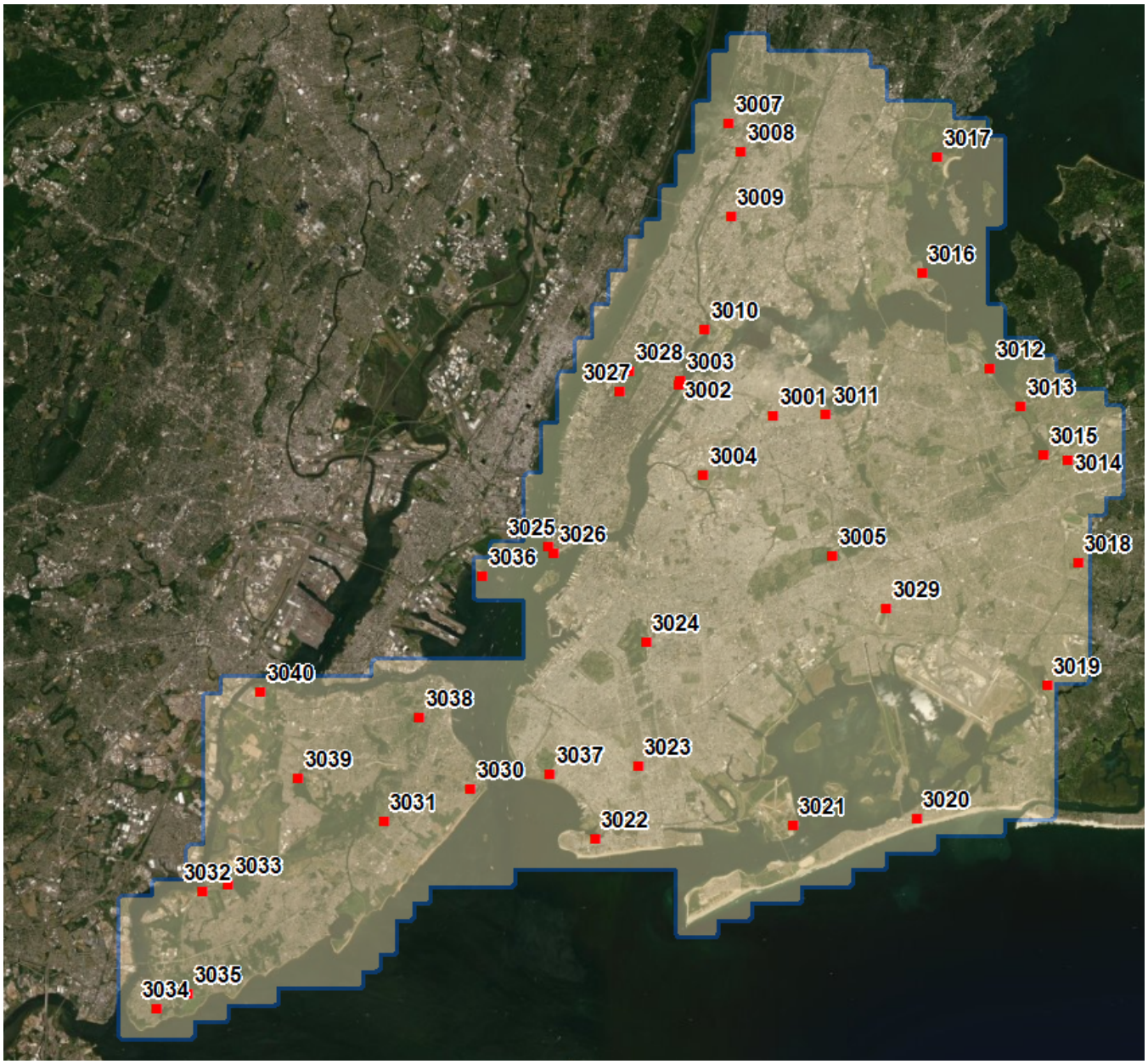
The following statistical results of the lidar data compared to the lidar control points post-calibration. The results show the difference between the lidar points and the 39 surveyed ground points.

Project Data Unit: U.S. Survey Foot
Vertical Accuracy Class tested: 10.0-cm
Elevation Calculation Method: Interpolated from TIN
LiDAR Classifications Included: 0-118, 120-255

Check Points in Report: 39
Check Points with LiDAR Coverage: 39
Check Points (NVA): 39
Check Points (VVA): 0
Average Vertical Error Reported: 0.000 U.S. Survey Foot
Maximum (highest) Vertical Error Reported: 0.301 U.S. Survey Foot
Median Vertical Error Reported: 0.000 U.S. Survey Foot
Minimum (lowest) Vertical Error Reported: -0.227 U.S. Survey Foot
Standard deviation of Vertical Error: 0.133 U.S. Survey Foot
Skewness of Vertical Error: 0.177
Kurtosis of Vertical Error: -0.399
Non-vegetated Vertical Accuracy (NVA) RMSE(z): 3.995cm PASS
Non-vegetated Vertical Accuracy (NVA) at the 95% Confidence Level +/-: 7.830cm PASS
FGDC/NSSDA Vertical Accuracy at the 95% Confidence Level +/-: 7.830cm
Non-vegetated Vertical Accuracy (NVA) RMSE(z) (DEM): 4.029cm PASS
Non-vegetated Vertical Accuracy (NVA) at the 95% Confidence Level +/- (DEM): 7.897cm PASS

This data set was tested to meet ASPRS Positional Accuracy Standard for Digital Geospatial Data (2014) for a 10.0-cm RMSEz Vertical Accuracy Class. Actual NVA accuracy was found to be RMSEz = 3.995cm, equating to +/- 7.830cm at the 95% confidence level.

Lidar Control Point Layout



Lidar Filtering and Classification

The lidar filtering process encompasses a series of automated and manual steps to classify the boresighted point cloud data set. Each project represents unique characteristics in terms of cultural features (urbanized vs. rural areas), terrain type and vegetation coverage. These characteristics are thoroughly evaluated at the onset of the project to ensure that the appropriate automated filters are applied and that subsequent manual filtering yields correctly classified data. Data is most often classified by ground and “unclassified”, but specific project applications can include a wide variety of classifications including but not limited to buildings, vegetation, power lines, etc. A variety of software packages are used for the auto-filtering, manual filtering and QC of the classified data.

Eagleview used the ASPRS LAS Specification Version 1.4 – R15 (ASPRS, 2011, published 09 July 2019), Point Data Record Format 6 for this project and classified the lidar point cloud in accordance with the following classification classes and bitflags. The following outlines project specific requirements.

- Class 1 = Unclassified
- Class 2 = Bare-earth Ground
- Class 3 = Low Vegetation
- Class 4 = Medium Vegetation
- Class 5 = High Vegetation
- Class 6 = Buildings (Automated)
- Class 7 = Low point (noise)
- Class 8= Model Key Point (Mass Point)
- Class 9 = Water
- Class 18 = High noise

Eagleview has developed several customized automated filters that are applied to the lidar data set based on project specifications, terrain, and vegetation characteristics. A filtering macro, which may contain one or more filtering algorithms, is executed to derive LAS files separated into the different classification groups as defined in the ASPRS classification table. The macros are tested in several portions of the project area to verify the appropriateness of the filters. Often, there is a combination of several filter macros that optimize the filtering based on the unique characteristics of the project. Automatic filtering generally yields a ground surface that is 85-90% valid, so additional editing (hand-filtering) is required to produce an accurate ground surface.

Lidar data is next taken into a graphic environment using MARS® to manually re-classify (or hand-filter) "noise" and other features that may remain in the ground classification after auto filter. A cross-section of the post auto-filtered surface is viewed to assist in the reclassification of non-ground data artifacts. The following is an example of re-classification of the non-ground points (elevated features) that need to be excluded from the true ground surface. Certain features such as berms, hilltops, cliffs and other features may have been aggressively auto-filtered and points will need to be re-classified into the ground classification. Data in the profile view displays non-ground (Unclassified, class 1) in grey and ground in brown/tan (Class 2). In **Figure 1**, a small building was not auto-filtered and needs to be manually re-classified. Note that **Figure 2** has the building points reclassified to unclassified from the true ground surface.

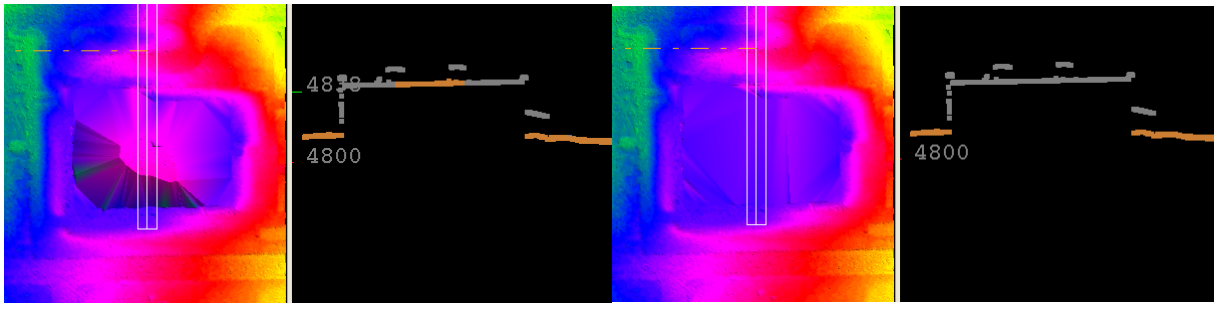


Figure 1

Figure 2

A combination of automated and semi-automated routines to classify buildings and vegetation. We expect that the classified buildings will meet expected filtering criteria.

At this point, individual lidar points from the original point cloud have now been parsed into separate classifications.

Filtered Lidar Checkpoint Report

After hand-filtering has been completed and quality checked, a Checkpoint Report is generated to validate that the accuracy of the ground surface is within the defined accuracy specifications. Each surveyed ground check point is compared to the lidar surface by interpolating an elevation from a Triangulated Irregular Network (TIN) of the surface. The MARS® derived report provides an in-depth statistical report, including an RMSE of the vertical errors; a primary component in most accuracy standards and a statistically valid assessment of the overall accuracy of the ground surface.

The below lidar check point reports provide statistics for 79 ground survey checkpoints (39 NVA, 40 VVA) used to validate the final filtered lidar surface.

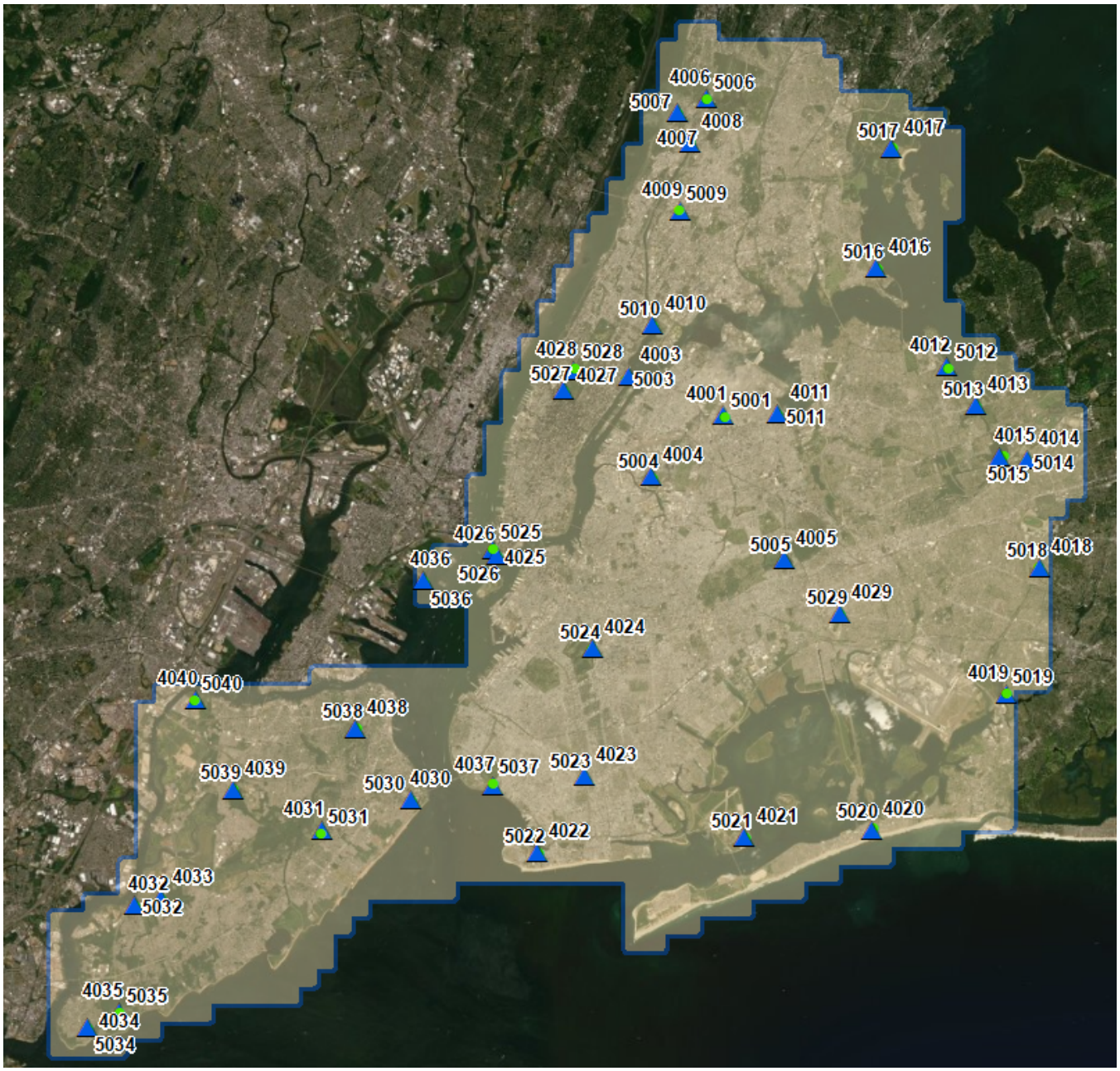
Project Data Unit: U.S. Survey Foot
Vertical Accuracy Class tested: 10.0-cm
Elevation Calculation Method: Interpolated from TIN
LiDAR Classifications Included: 2,8/0 Ground (All)/0W

Check Points in Report: 79
Check Points with LiDAR Coverage: 79
Check Points (NVA): 39
Check Points (VVA): 40
Average Vertical Error Reported: -0.026 U.S. Survey Foot
Maximum (highest) Vertical Error Reported: 0.246 U.S. Survey Foot
Median Vertical Error Reported: -0.022 U.S. Survey Foot
Minimum (lowest) Vertical Error Reported: -0.299 U.S. Survey Foot
Standard deviation of Vertical Error: 0.125 U.S. Survey Foot
Skewness of Vertical Error: -0.023
Kurtosis of Vertical Error: -0.404
Non-vegetated Vertical Accuracy (NVA) RMSE(z): 3.846cm PASS
Non-vegetated Vertical Accuracy (NVA) at the 95% Confidence Level +/-: 7.539cm PASS
Vegetated Vertical Accuracy (VVA) at the 95th Percentile +/-: 39.432cm FAIL
FGDC/NSSDA Vertical Accuracy at the 95% Confidence Level +/-: 7.539cm
Non-vegetated Vertical Accuracy (NVA) RMSE(z) (DEM): 3.577cm PASS
Non-vegetated Vertical Accuracy (NVA) at the 95% Confidence Level +/- (DEM): 7.012cm PASS
Vegetated Vertical Accuracy (VVA) at the 95th Percentile +/- (DEM): 39.231cm FAIL

This data set was tested to meet ASPRS Positional Accuracy Standard for Digital Geospatial Data (2014) for a 10.0-cm RMSEz Vertical Accuracy Class. Actual NVA accuracy was found to be RMSEz = 3.846cm, equating to +/- 7.539cm at the 95% confidence level. Actual VVA accuracy was fo und to be +/- 39.432cm at the 95th percentile.

Lidar Checkpoint Layout

NVA
VVA



List of Deliverables

- Softcopy of the flight plan
- GCPs in shapefile format
- Post calibration / Pre-Classified point cloud in LAS 1.4 format
- Fully classified LiDAR point cloud in LAS 1.4 format
- LiDAR Collection, Processing and accuracy report

Appendix 1

Following is a more detailed lidar calibration workflow description.

LIDAR CALIBRATION AND BLOCK LAS OUTPUT

Note: All figures represented on the following pages are for general illustration purposes, and are not examples derived from the project.

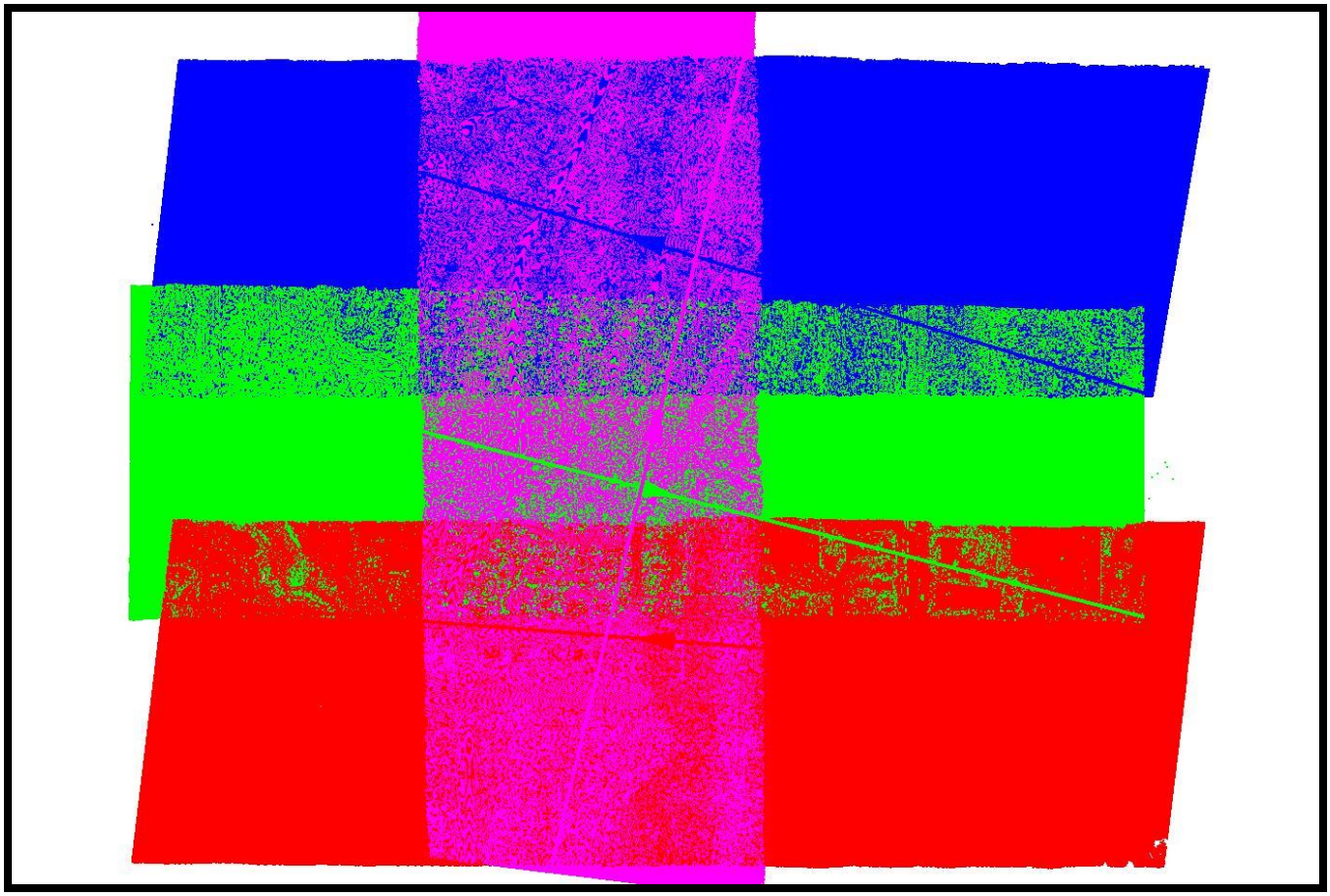
Initial Processing

Lidar data is output as LAS point data using Optech's Lidar Mapping Suite (LMS). LMS matches ground and roof planes plus roof lines to self-calibrate and correct system biases. These biases occur within the hardware of the laser scanning systems, within the Inertial Measurement Unit (IMU) and because of environmental conditions which affect the refraction of light. The systemic biases that are corrected for include scale, roll, pitch, and heading.

In addition to the self-calibration mode LMS runs a "production" mode which applies the self-calibration parameters and then analyzes each individual flight line and applies small adjustments to each line to tie overlapping lidar points even more tightly together.

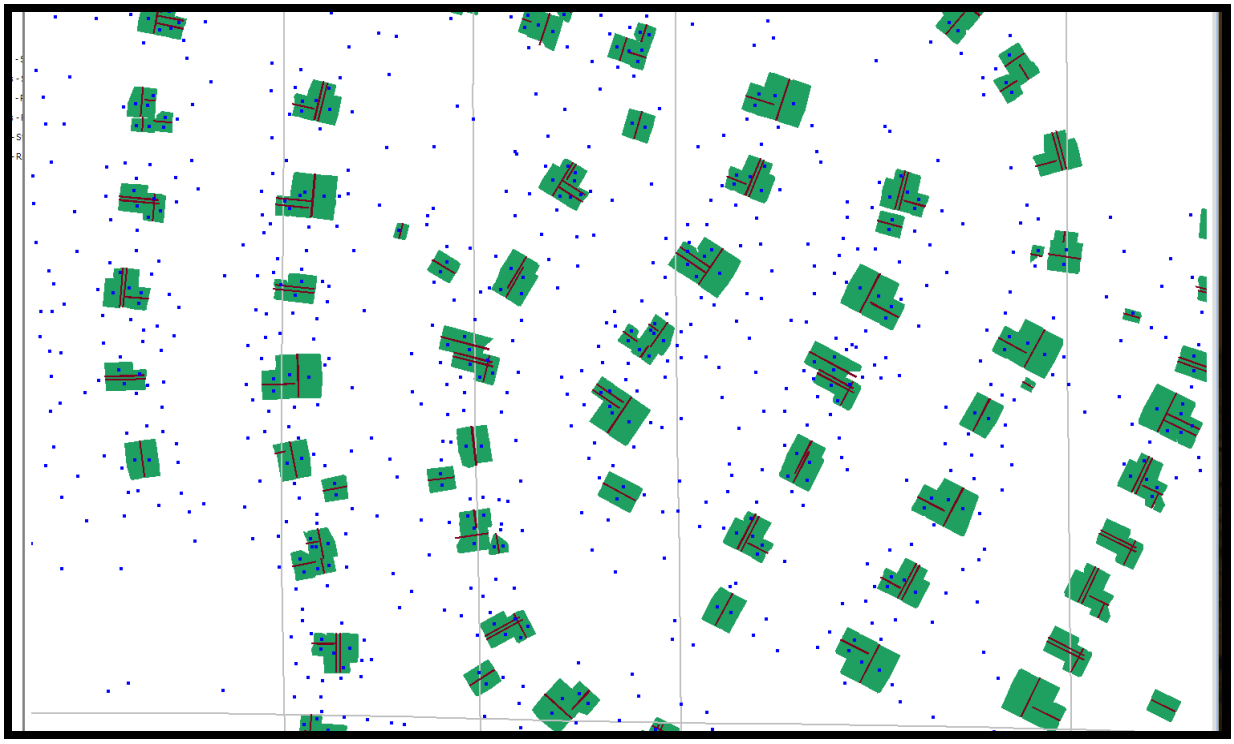
Boresight Self-Calibration Processing Procedures

An LMS boresight calibration is performed on an as-needed basis to correct scale, roll, pitch and heading biases. A minimum of three overlapping flights are flown in opposing directions with one cross flight.



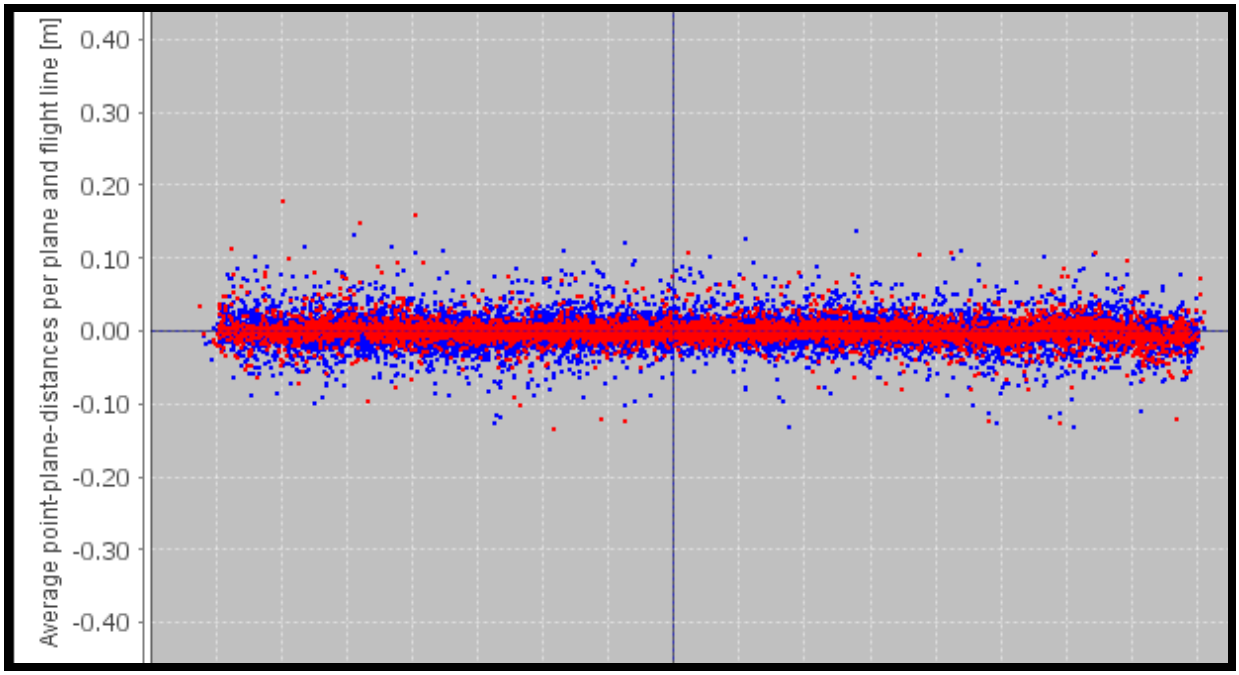
The Boresighting module frees scan angle scale, scan angle lag, XYZ boresight corrections and elevation position corrections while locking scan angle offset and XY position corrections.

The picked calibration site will have a good distribution of buildings for the self-calibration software to match ground planes, roof planes and roof lines.

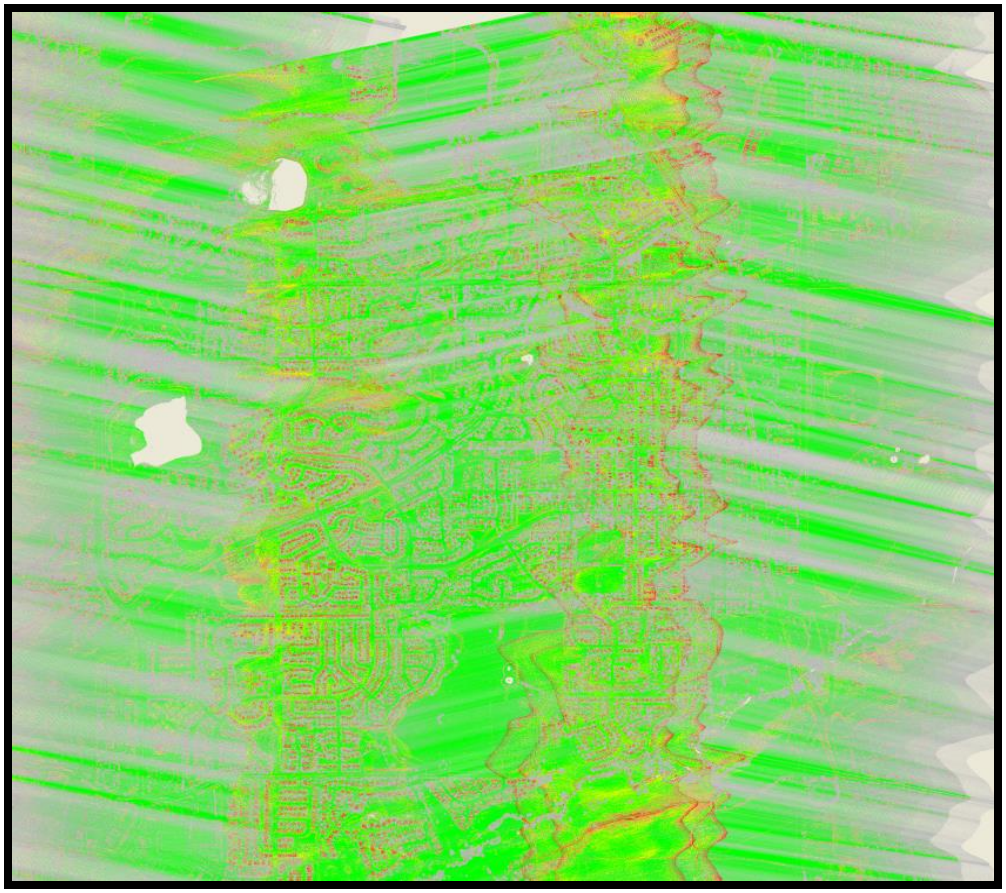


At the conclusion of the self-calibration run the data is quality checked with LMS plots

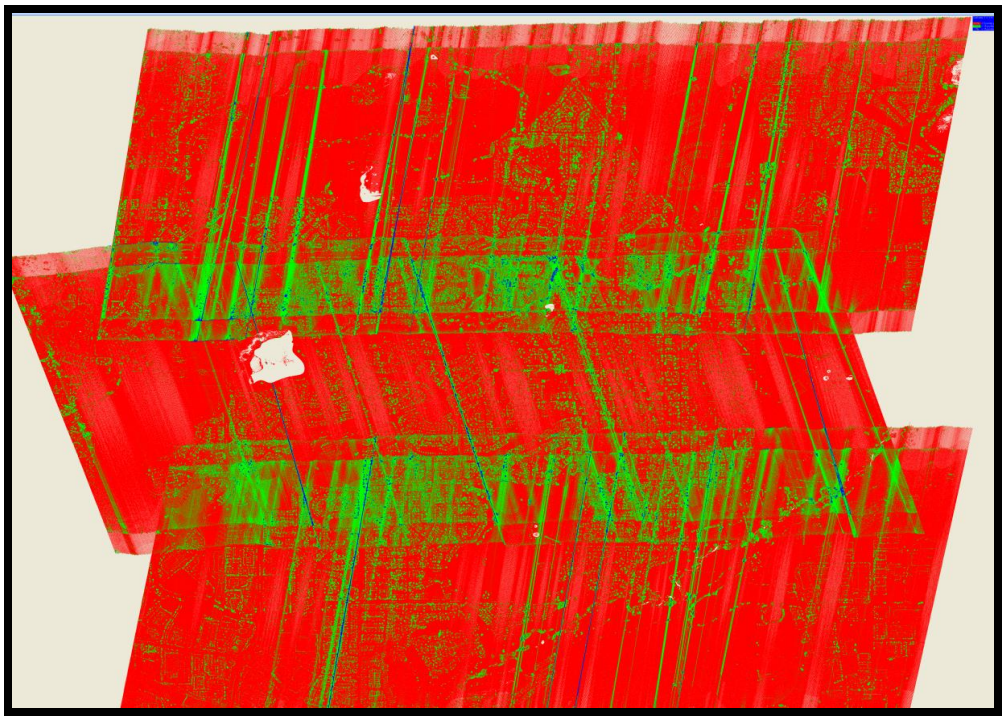
Plot of plane vertical distances from datum plane.



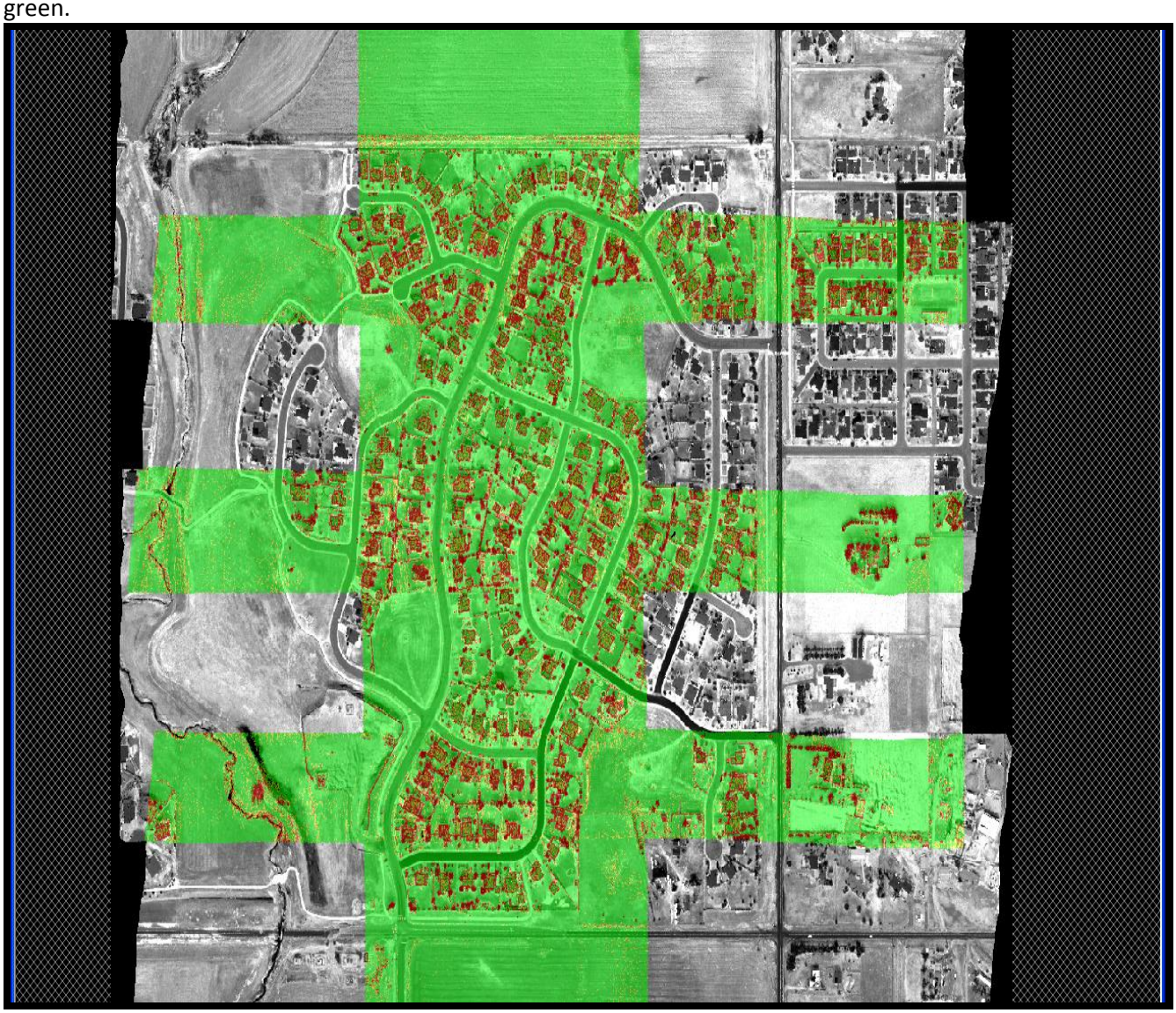
Plot of height differenced between flight lines. (Green=less than 5cm).



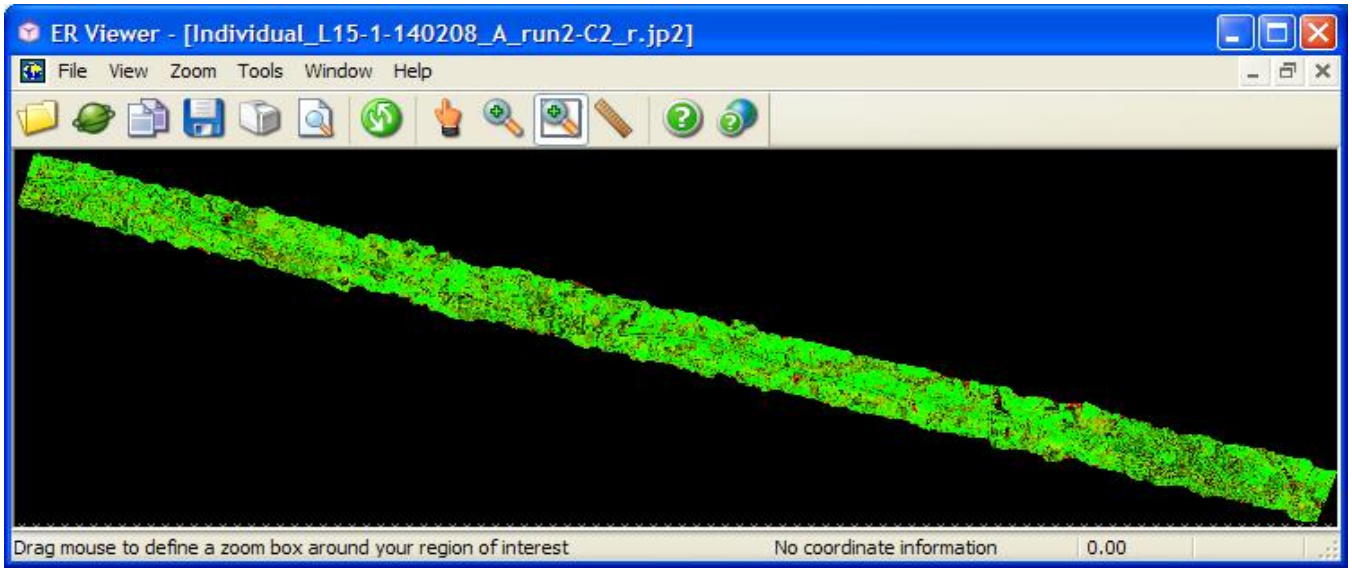
Plot of point densities. (Red=5-9 points per cell, green 10+ points per cell).



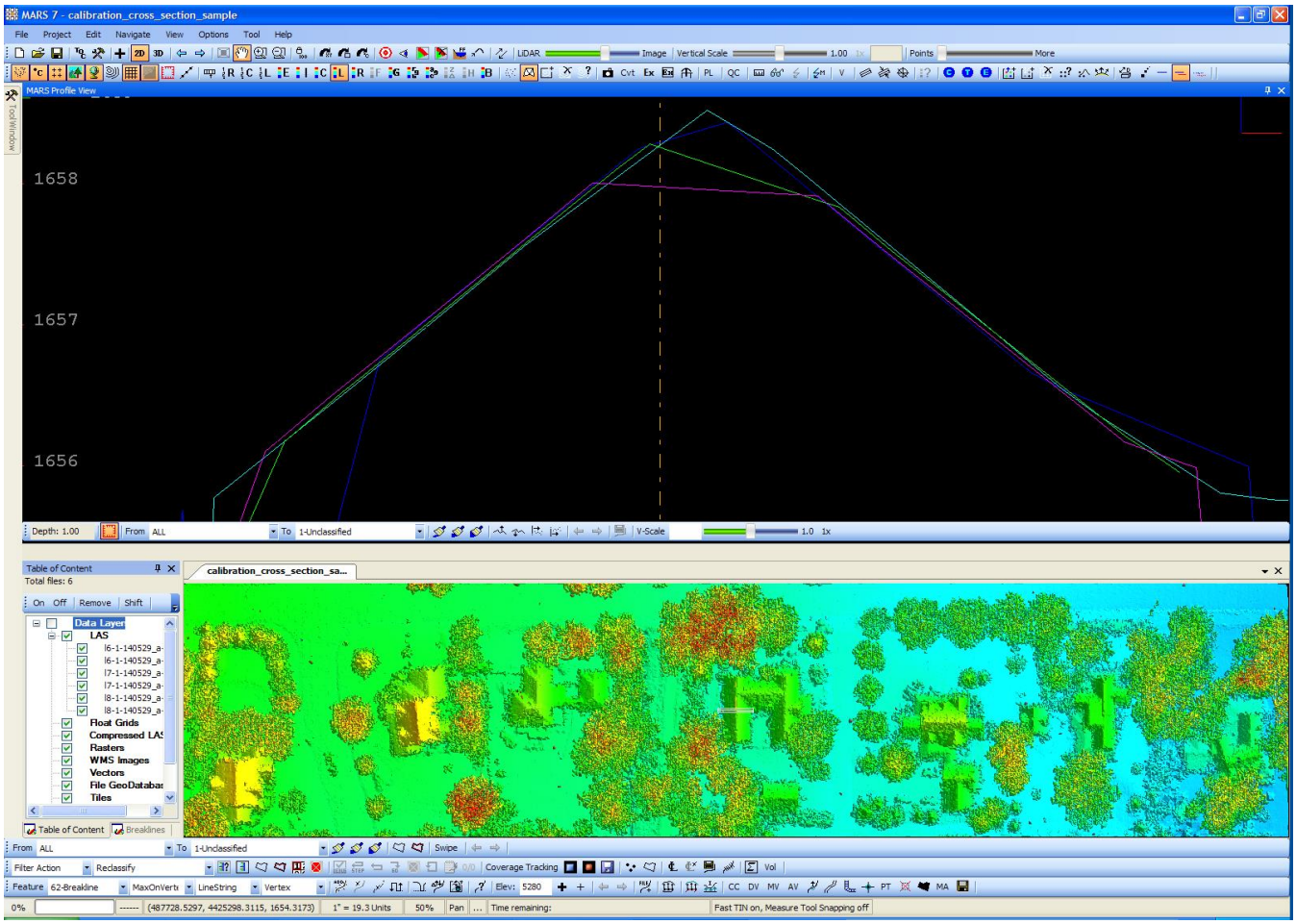
A Flight Line Separation Raster image is generated in Eagleview Advanced Remote Sensing Software (MARS®), in this example ground returns from multiple flight lines that are fitting within 3 centimeters are colored green.



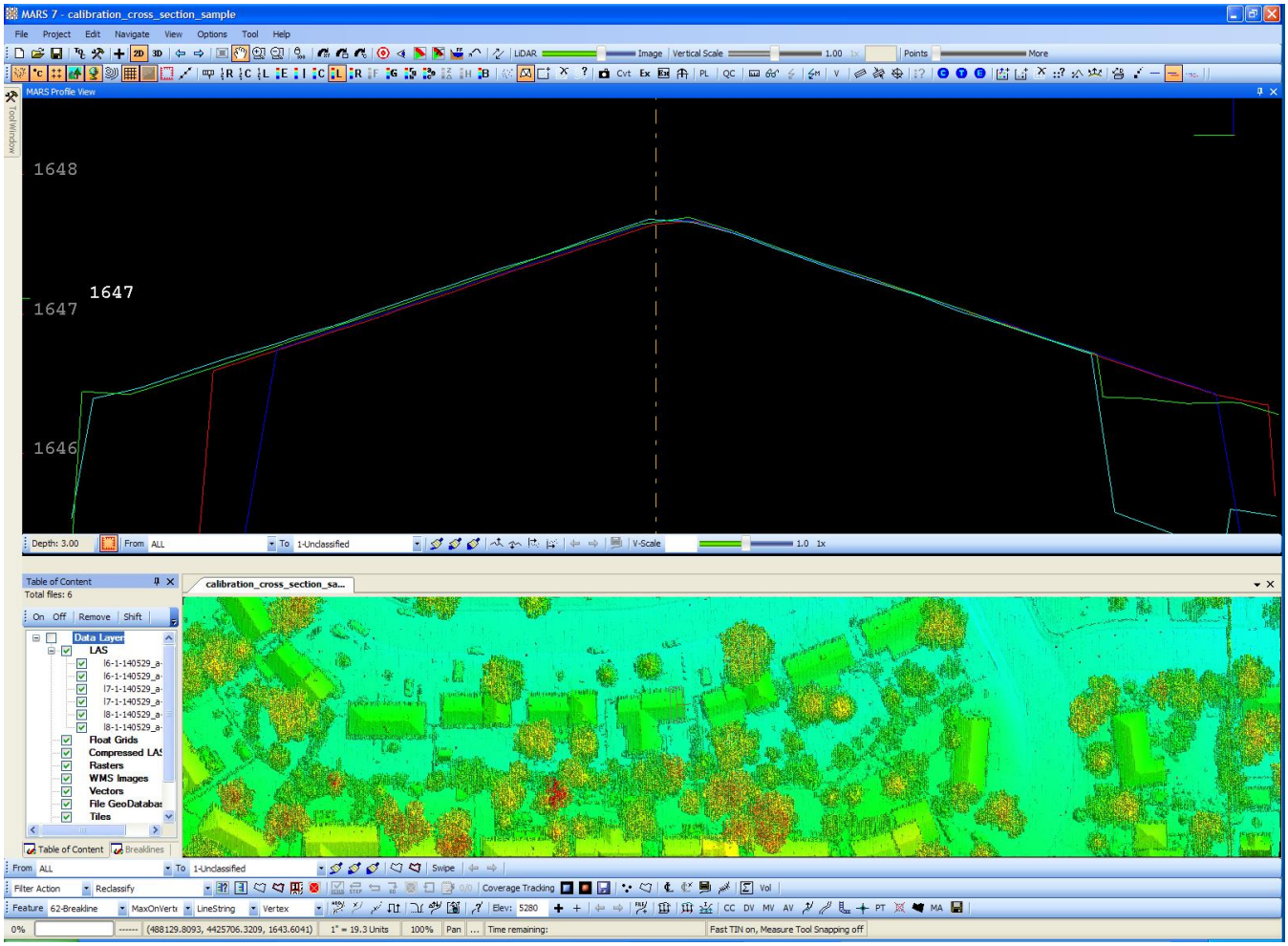
MARS® tests for internal relative vertical accuracy using inbound and outbound scan values. Again, Green is showing inbound and outbound scan data fitting to 3 centimeters.



Building cross sections are checked for good alignment. Pitch and heading are checked on roof planes parallel to the flight direction.



Roll and scale are checked on roof planes perpendicular to the flight direction.

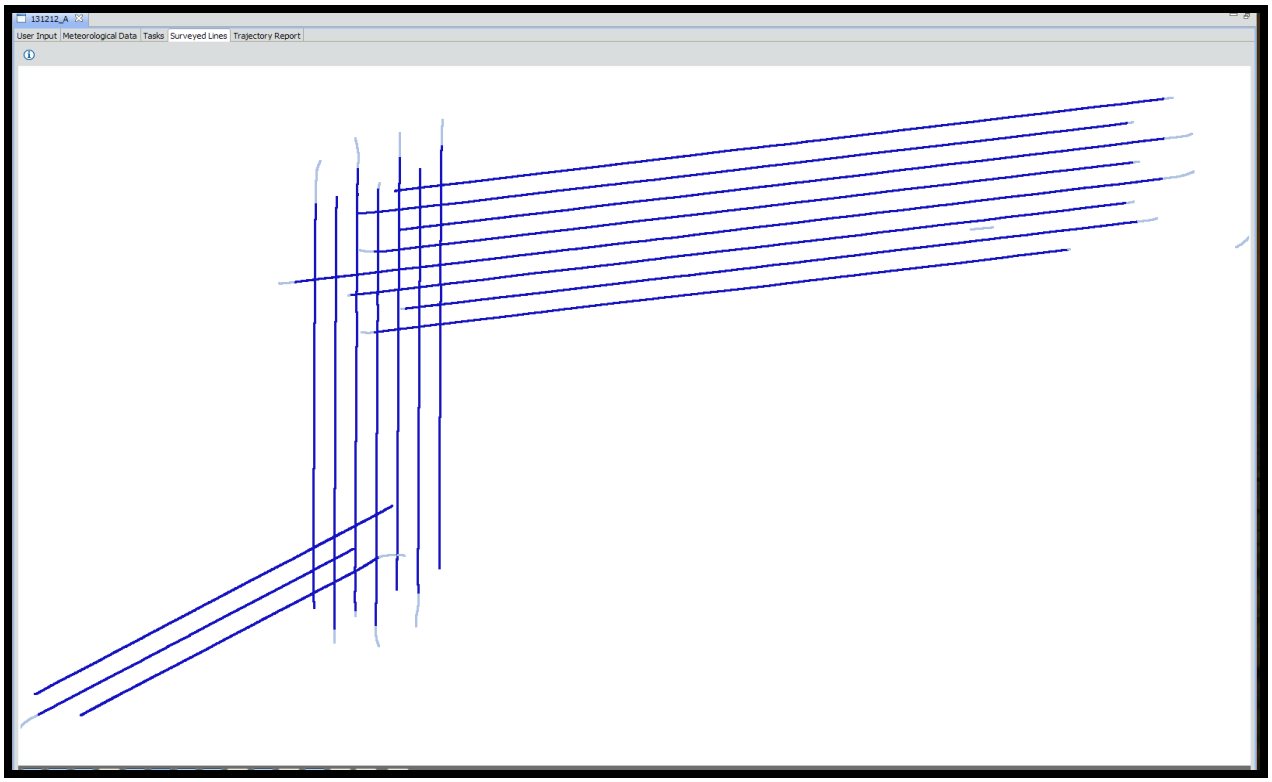


The LMS program outputs a "LCP" file with all the correction parameters. The calibration process may be run several times until the boresight adjustments are acceptable. When the boresight solution is acceptable the LCP file adjustments are saved and also applied to subsequent projects. Each new project is again analyzed and when the adjustment biases show too much drift a new boresight calibration is run. The LCP file may hold calibration tolerances for several projects.

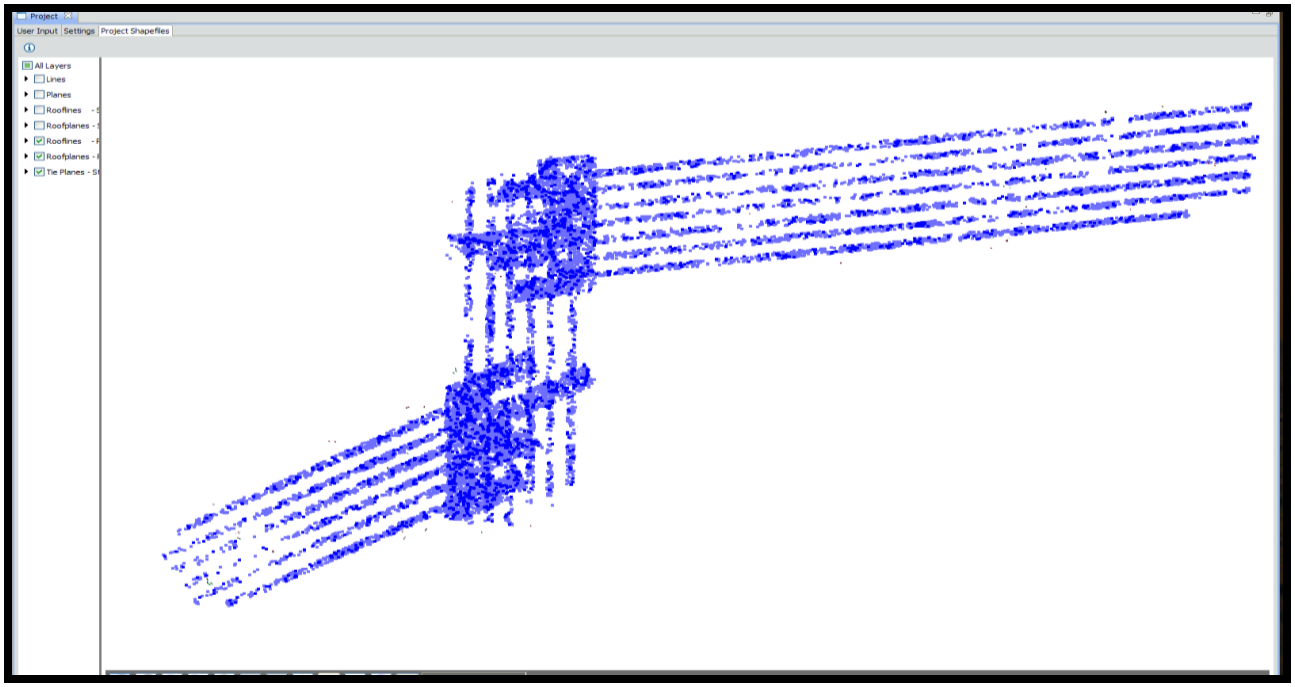
Block LAS Production Processing Procedures

The LMS production mode is run on each flight line to further tie the final lidar LAS flight line files tightly together. Production settings allow scan angle scale, scan angle lag to float and allows elevation to move slightly during flight line to flight line comparison thus further tying flight lines together. A cross flight with locked elevation data is used for controlling flight line elevations.

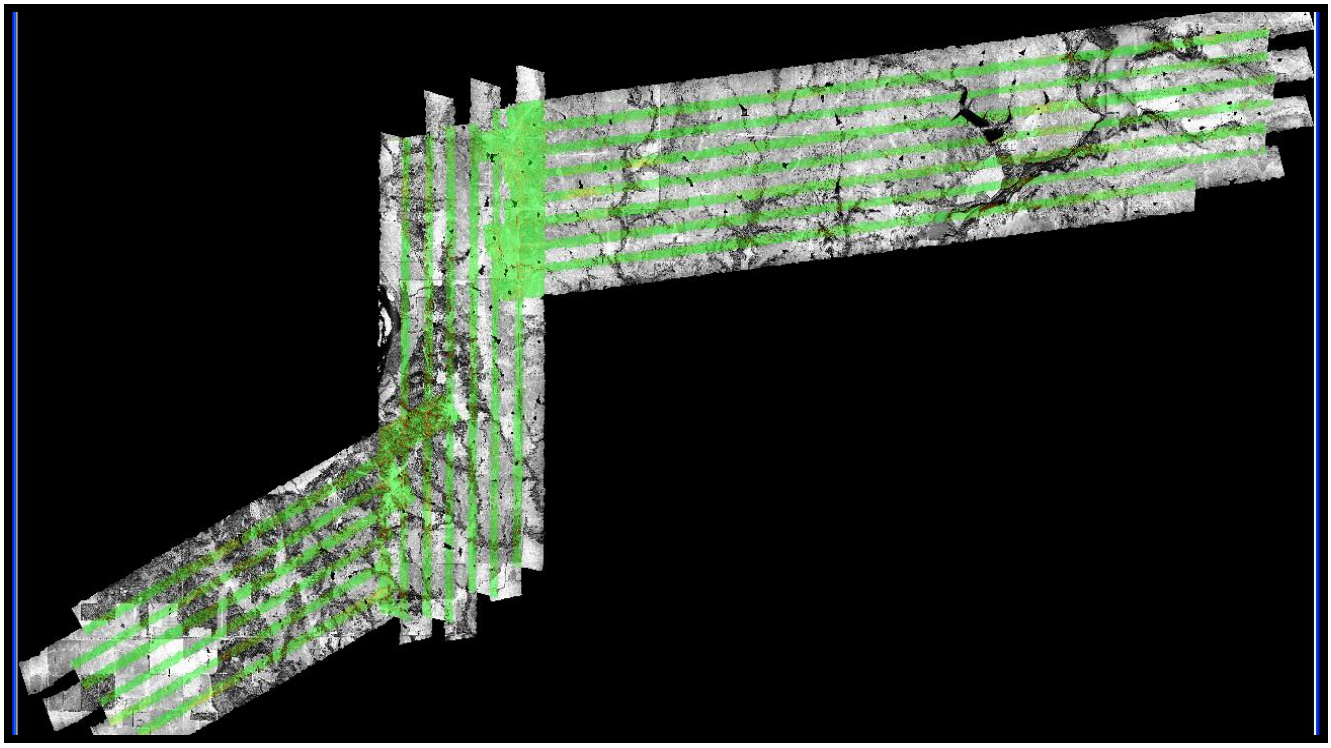
A block of data is selected to process with LMS production settings. Data collected during turns at the ends of flight lines is deselected (light blue lines).



As in self-calibration the LMS production program analyses ground, roof planes and rooflines. One cross flight is locked in elevation and all other lines are adjusted to it. Unlike the calibration site the distribution of roof planes is usually much less dense. Here matched ground tie planes are blue.

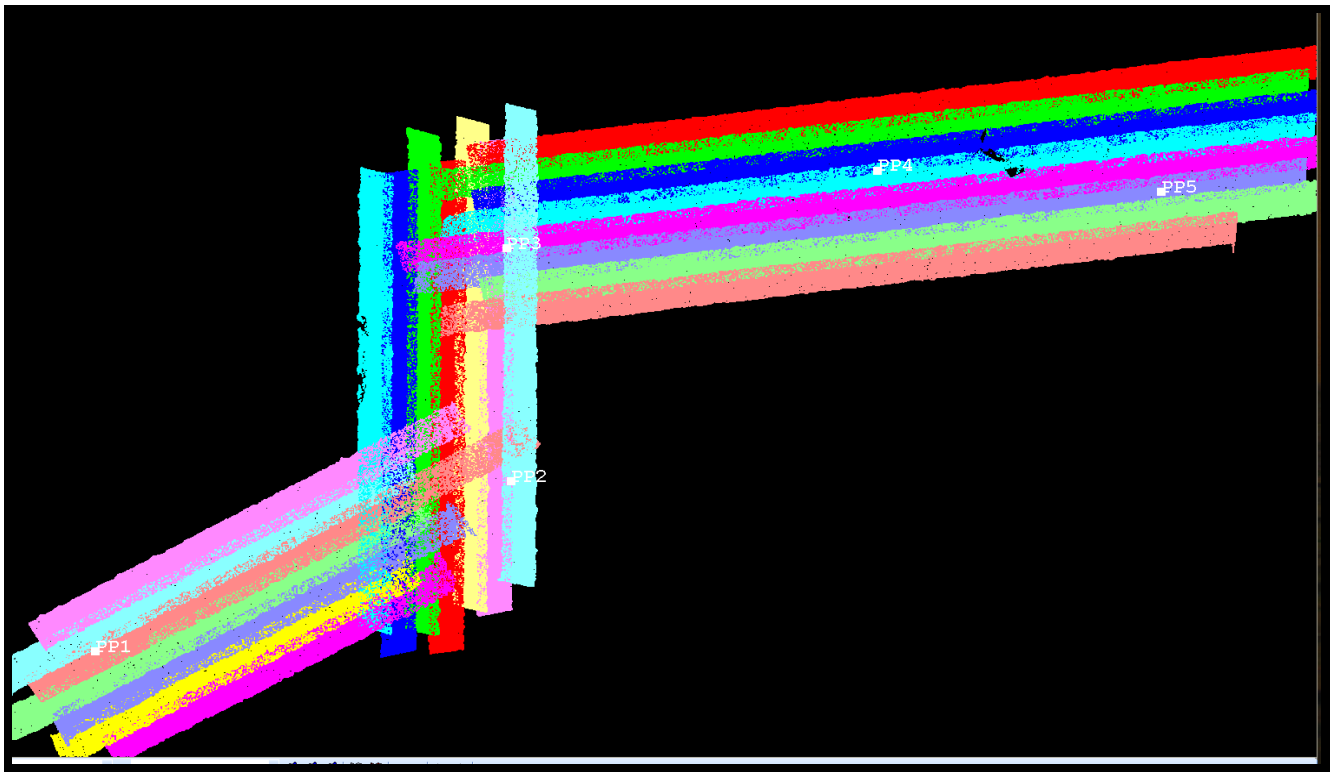


The same quality control outputs used to check self-calibrations are available to analyze the production run. Output plots are again available in LMS and cross sections plus a Flight Line Separation Raster are generated in MARS® to check coverage and quality.



Correcting the Final Elevation

After all the lines are tied together a ground control network is imported into MARS®. The ground control network may be pre-existing or collected by a licensed surveyor.



The next step is to match the ground control elevations to the lidar data set. A control report is run and the data set is shifted slightly to zero out the average elevation error and points checked for quality.

The final step before boresighted, leveled LAS files are ready for filtering is to run the MARS® QC Module on the block data. The Boresighted lidar QC Report outputs individual reports on Point Density, Nominal Pulse Spacing, Data Voids, Spatial Distribution, Scan Angles, Control Report, Flight Line Separation, Flight Line Overlap, Buffered Boundary, LAS Formats, Datums and Coordinates.
These reports are checked with the required specifications in the Project Management Plan.